Course Project : Statistics Portal

Name : Sayed Al Sami

Student ID : 003055997

Introduction

Finland Data Explorer is a web application that allows users to explore population, employment and unemployment statistics across Finnish municipalities. The purpose of the project is to make statistical information easier to understand through interactive maps, charts and data analysis tools.

The application uses HTML, CSS and JavaScript. It retrieves statistical data from Statistics Finland and displays the information in a visual format.

Data Sources

The application uses Statistics Finland as its main source of statistical information. Population data and labour-market statistics are used to compare Finnish municipalities. Geographic boundary data is used to display the municipalities on an interactive map. OpenStreetMap provides the background map.

The data sources are linked in the application so users can find more information about them.

Implementation and Features

The application combines interactive maps, charts and statistical analysis in one interface. Users can select different indicators to explore population and labour-market statistics. The map uses colours to represent the selected data, while the chart provides another way to compare municipalities.

The data analysis tools allow users to calculate statistical summaries and identify municipalities with the highest values. Additional functionality includes drag-and-drop interactions, mathematical operations between datasets and exporting supported visualizations. These features help users explore the data in different ways.

Technologies

HTML was used to structure the application, while CSS was used to create the layout and responsive styling. JavaScript implements the interactive features and processes the statistical data. Leaflet is used to display the map, and Frappe Charts is used to create charts. Statistical information is retrieved from Statistics Finland's data services.

Testing

The application was tested by checking its main interactive features, including changing map and chart indicators, performing statistical calculations and exporting visualizations. The responsive layout was also checked at a mobile screen width.

The results were recorded to identify features that work correctly and areas that require further testing. Browser compatibility and the availability of live external data sources must also be verified before the application is considered ready for submission.

Conclusion

Finland Data Explorer demonstrates how web technologies can be used to present public statistical data interactively. The application combines maps, charts and data analysis tools to make information about Finnish municipalities easier to explore.

The project also provided practical experience with JavaScript, external APIs, data visualization and interactive user interfaces. Further improvements could include additional datasets, more advanced comparisons and more extensive testing across browsers.

Use of AI Tools

AI assistance (ChatGPT) was used to support code development, explain programming concepts, identify potential errors and help prepare project documentation. The application was tested to the extent described in this report, and the final implementation and test results were reviewed before submission.